

Assignment - 7

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Problem - 1:

Below compare the capability of the output for the different face detection YOLO versions

Parameter	YOLOv8	YOLOv11	YOLOv12
Processed 1081 frames	17.47 seconds	19.36 seconds	21.87 seconds
Average FPS	61.89	55.83	49.43
Inference time	0.010 seconds	0.011 seconds	0.014 seconds
Detected Objec	bird: 17 cake: 11 motorcycle: 11 person: 2642 car: 85 clock: 8 boat: 16 handbag: 44 tie: 2 snowboard: 4 bottle: 4 skateboard: 7 carrot: 4 airplane: 67 bed: 1 cell phone: 8 suitcase: 4 umbrella: 14 bus: 37 train: 15	pizza: 2 bird: 11 person: 2444 motorcycle: 35 car: 149 airplane: 110 umbrella: 30 handbag: 36 sports ball: 4 cell phone: 24 suitcase: 4 backpack: 18 carrot: 17 train: 67 parking meter: 8 boat: 9 bus: 6 hot dog: 2	bird: 27 airplane: 85 clock: 2 person: 2424 car: 254 motorcycle: 19 bus: 2 truck: 1 cell phone: 1 backpack: 18 teddy bear: 9 sports ball: 6 tie: 10 handbag: 21 suitcase: 8 carrot: 27 keyboard: 2 kite: 3 bed: 1 ... vase: 1 train: 9 dog: 1 boat: 3

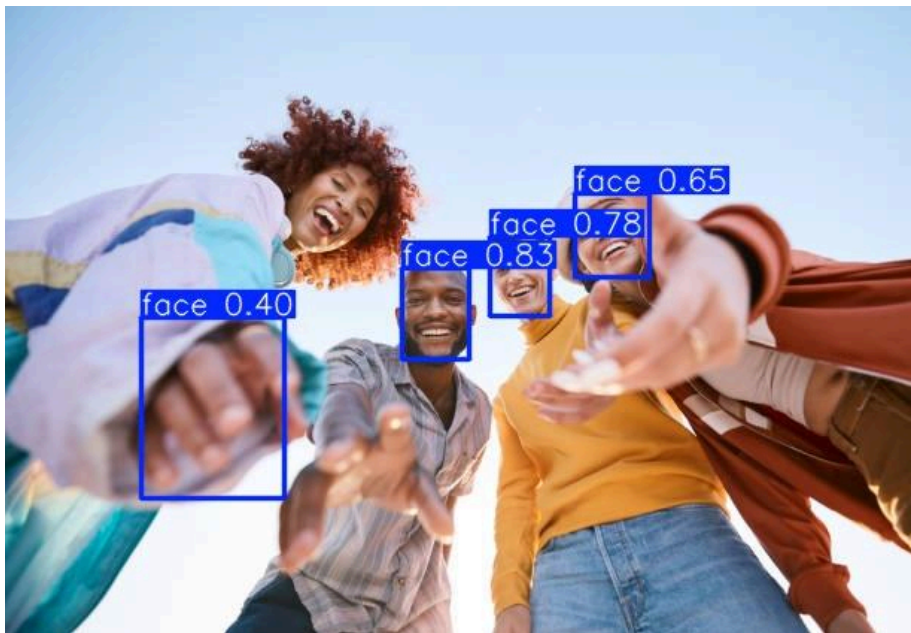
Problem - 2:

Below given the github code link for fine tuned the YOLOv8 model and splitting the WIDER-Face dataset and the output for the trained model.

Input image:



Output image:



Problem-3:

Below given the comparable for the Ultralytics YOLOv8 model and the given github link
YOLOv8 model

Input image for the two models are:



Output for the Ultralytics YOLOv8 model is



Output for the given github linked model:



Here we can see that the Ultralytics YOLOv8 model were not performed better than the given github code link. The given linked model are detect more faces than the Ultralytics model.

Problem - 4:

Below given the YOLOv1 model code link and which is trained and validate by the WIDER-Face dataset and the output for this model.

The mode code : YOLOv1

Output:

